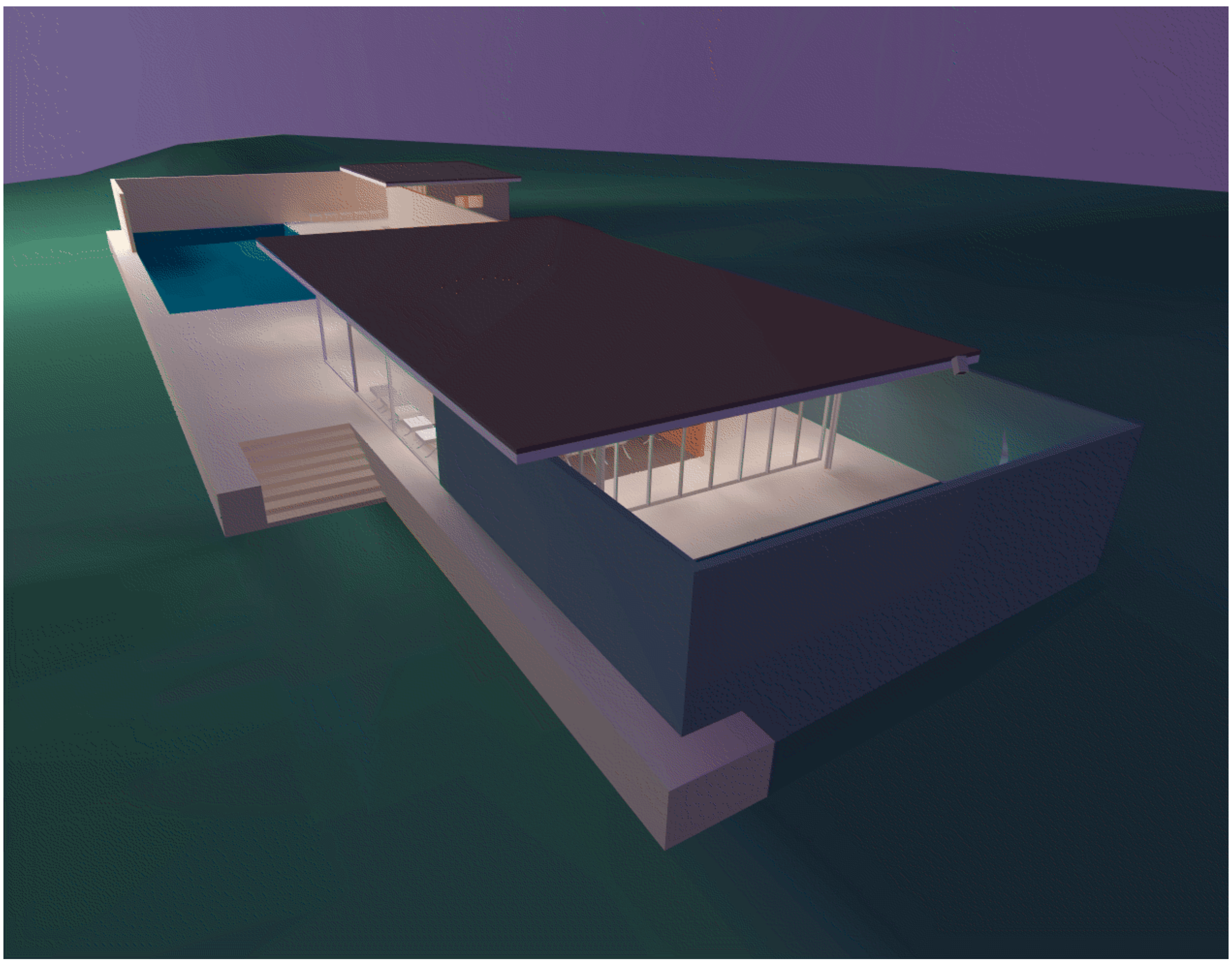
SGI GFO Format

The GFO format is the simple ASCII format of the barcelona database that is provided in the OpenGL Performer sample database directory. This database represents the famous German Pavilion at the Barcelona Exhibition of 1929, which was designed by Ludwig Mies van der Rohe and is shown in Figure 7-5.

Figure 7-5. GFO Database of Mies van der Rohe's German Pavilion



The source code for the GFO-format loader is provided in the file /usr/share/Performer/src/lib/libpfdb/libpfgfo/pfgfo.c.

pfdLoadFile() uses the function **pfdLoadFile_gfo()** to load GFO format files into OpenGL Performer run-time data-structures.

When working with GFO files, remember that hardware lighting is not used since all illumination effects have already been accounted for with the ambient color at each vertex.

The GFO format defines polygons with a color at every vertex. It is the output format of an early radiosity system. Files in this format have a simple ASCII structure, as indicated by the following abbreviated GFO file:

```
scope {
v3f {42.9632 8.7500 0.9374}
cpack {0x8785a9}
v3f {42.9632 8.0000 0.9374}
cpack {0x8785a9}
...
v3f {-1.0000 -6.5858 10.0000}
cpack {0xffffff}
polygon {cpack[0] v3f[0] cpack[1] v3f[1] cpack[2] v3f[2] cpack[3] v3f[3] }
polygon {cpack[4] v3f[4] cpack[5] v3f[5] cpack[6] v3f[6] cpack[7] v3f[7] }
...
polygon {cpack[7330] v3f[7330] cpack[7331] v3f[7331] cpack[7332] v3f[7332] cpack[7333]
v3f[7333] }
instance {
polygon[0]
polygon[1]
...
polygon[2675]
}
}
```

This example is taken from the file barcelona-l.gfo, one of only two known databases in the GFO format. The importer uses functions from the libpfdu library (such as those from the pfdBuilder) to generate efficient shared triangle strips. This increases the speed with which GFO databases can be drawn and reduces the size and complexity of the loader, since the builder's functions hide the details of the pfGeoSet construction process.